

Data Storytelling



Module 1: Data Storytelling Fundamentals

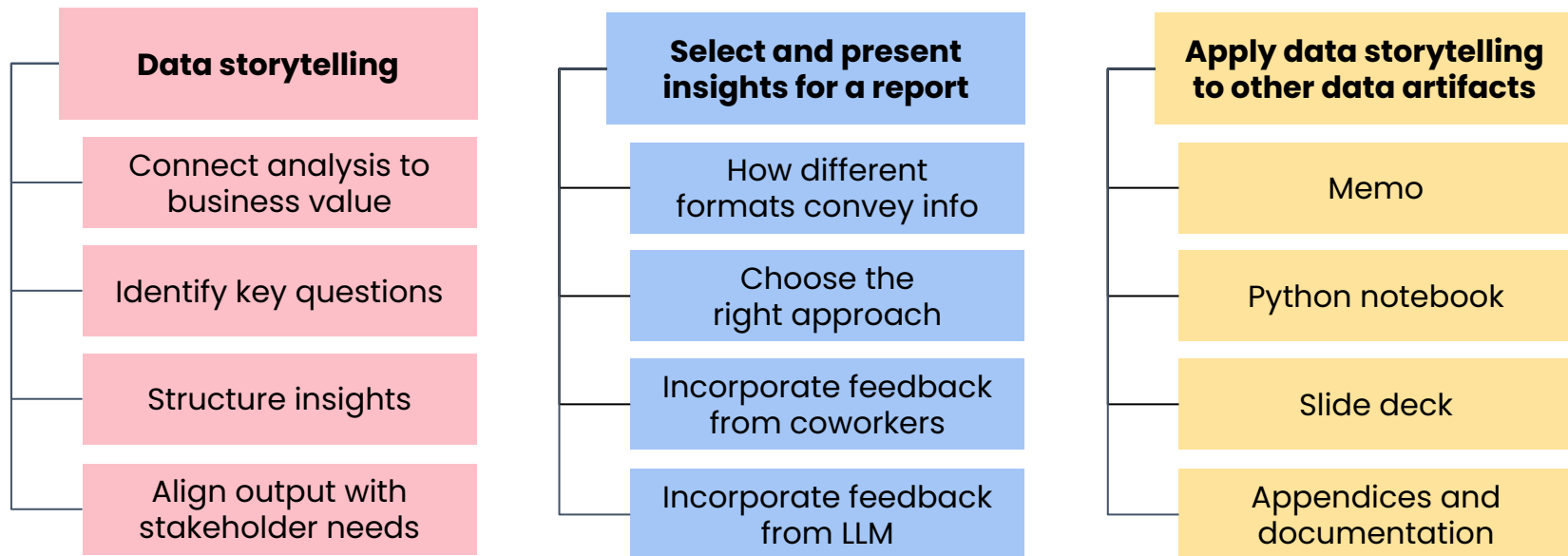




Data Storytelling Fundamentals

Module 1 introduction

Module 1 outline





Data Storytelling Fundamentals

From technical skills to
business value

Revisiting the lifecycle

✉ **Manager:** “Hey, there’s been an increase in customer engagement lately, can you look into it?”

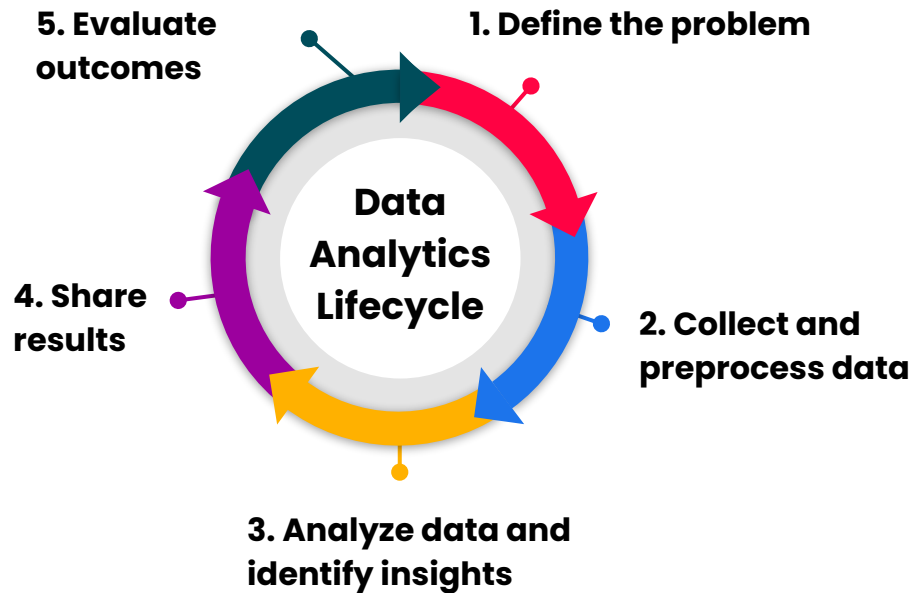
✗ **Less insightful answer:** “Engagement increased by 5% last month”

What manager is asking: a **story**:

- Why are customers engaging more?
- How unusual is this trend?
- What can the business do to keep it going?

💬 Data storytelling begins when defining the problem:

- Start developing a narrative about **what** happened and **why**.



Connecting data to business context

- Consider the best way to share the insights
- Think about:
 - Who your audience is
 - What decisions they're trying to make
- **Example:** Active users grew 5% last month
 - **Executives:** Contextualize how current trend compares with previous months.
 - **Product managers:** Segmentation analysis of how engagement increased among different user groups
 - **Data team:** Seek feedback on analytical approach and get help interpreting insights

Stakeholder	Wants to know
Executives	• How insights impact revenue, strategy, or growth
Product managers	• User behavior to improve features • Case studies • User profiles
Members of the data team	• Which analyses you chose and why

Turning analysis into insights

- Depending on audience, you'll decide on the best format to present your results
- **Example:** Data table
 - Might contain wealth of statistics
 - Key figures may get buried in a mountain of less important numbers
- Insights should be:
 - ✓ Clear
 - ✓ Actionable
- **Example:** Presenting to executives
 - Short slide deck
 - Share:
 - **Key metric:** Active users grew by 5% last month
 - **Reason:** Price of basic tier was lowered last month
 - **Visualization:** Line chart of active users over time
 - Impact of overall revenue



Data Storytelling Fundamentals

Data storytelling

What is data storytelling?

Data storytelling:

- Shaping analysis into narrative that guides audience toward understanding and action
- Ensures insights lead to real-world decisions

Traditional report: Table of monthly sales figures with subscription dates, user locations, shows watched, etc.



- **Role:** User experience designer
- **Task:** Well-designed app makes it easy for users to accomplish goal

Example: Tracking sales performance



When crafting a data story:

- Act as a tour guide through key insights
- Make it easy to get key takeaways

Data story:



Q3 sales dropped because of supply chain disruptions



Customer retention increased due to lowered prices



Targeted promotions helped regain lost customers

Artifacts



Slide deck



Memo



Notebook



Dashboard

- Consider the visual design and layout to guide your audience through the artifact
- Common considerations:
 - Selecting a main conclusion
 - Supporting evidence
 - Ordering evidence

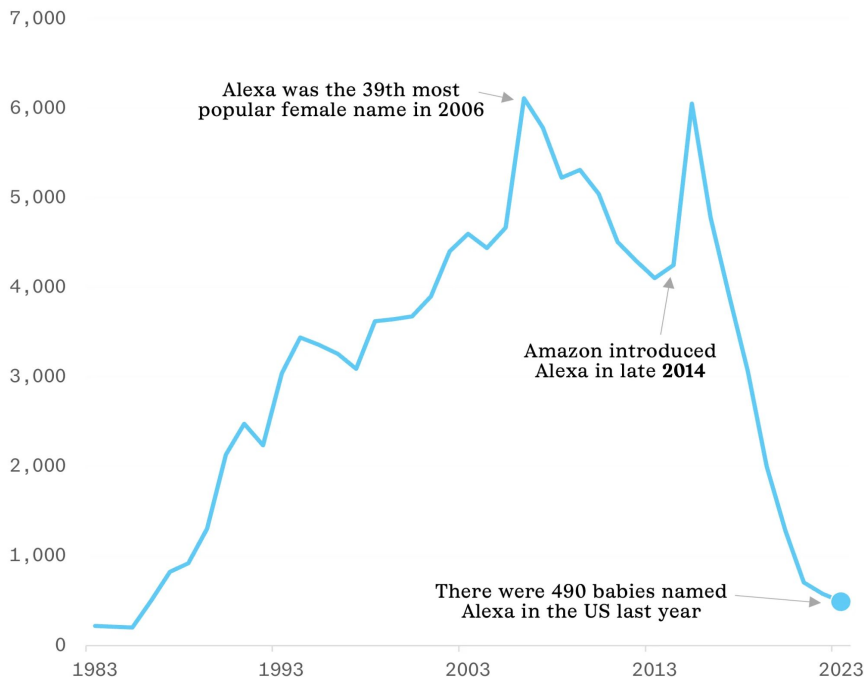
Power of visualization

- Good data visualization is the cornerstone of data storytelling
- Well-designed chart communicates insights more effectively than raw numbers

- ✓ Line chart is best choice for this data
- ✓ Minimizes complexity
- ✓ Uses clear text annotations
- ✓ Has a consistent color scheme
- ✓ This visualization:
 - Can support main narrative
 - Clearly communicates likely reason for decrease in babies named Alexa

Parents Have Stopped Calling Kids Alexa

Number of Babies Named Alexa [US, female, by birth year]



Source: Social Security Administration

[Chartr.co., The rise and fall of the name Alexa (2024)]

Data storytelling process

1 Start with a question or problem

- Before analysis, define the decision.
 - Why have horror movies seen a pronounced increase in viewership?
 - Which segment of users is driving the most revenue?

2 Choose the right data

- Focus on the data that provides insight into your key question.
- **Goal:** Understand horror movie viewership
 - ✓ **Helpful:** Viewership by genre & country
 - ✗ **Less helpful:** Subscription count by tier

3 Perform analysis

- Find patterns, trends, and relationships in the data.
- Look for:
 - Anomalies
 - Significant changes
 - External market forces
 - New initiatives by your company

4 Develop storytelling artifact

- Package your findings to inspire action
 - Creating a single great visualization
 - Writing a memo



Data Storytelling Fundamentals

Crafting a narrative

Crafting a narrative



Analogy: Essay writing in school



Basic structure:

- Introduction
- Three paragraphs of details
- Conclusion



In college or university:

- Literature review
- Methodology
- Conclusion



As a data analyst:



Throw these models out!



Stakeholders don't have time
this setup before main point

Understanding your audience

- Data storytelling involves **perspective taking**

👁️ Seeing situation from someone else's viewpoint

- Different audiences care about different things

🤝 Get to know your stakeholders:

- What challenges are they facing?
- What decisions do they need to make?
- What kind of information helps them do their job?

- A strong data narrative presents insights objectively, even if challenging



Remember: You are a steward of truth

Structure your data story



Determine audience's goals



Structure your story:

1. Start with **main conclusion**



Answer to main **business question**

2. Explain **insights** that support
main conclusion



Details about **data** used to draw
that conclusion

3. Revisit your **main conclusion**
○ Provide recommendations
on next steps

○ Context, historical
information, sample, etc.



Explain supporting **evidence**

Example

1. Begin with most important insight

"Horror movie viewership **has increased by 17% over the past month**, the **largest** month over month increase ever. This viewership increased **due to sponsorship deals** with top horror game streamers. By **expanding the sponsorship program**, we can see similar increases across other genres."

2. Present analysis used to reach conclusion

"We analyzed the behavior of over **30,000 users**, including both first-time horror movie viewers, as well as users who had shown an interest in the genre before. Our aim was to **predict user behavior** across all genres. We analyzed user behavior on our site after clicking on sponsored links, including **clicks and watch time**."

3. Explain how each insight supports conclusion

"We analyzed **watch time for the first horror movie ever viewed** by a user. If the user discovered that movie by clicking on a sponsored link, their **average watch time was 17% higher** compared with users who discovered the movie through another avenue, like their homepage or discovery view. This **difference was statistically significant at the 1% level**, so we have high confidence that this is a true effect. Here's a graph of watch time..."

4. Restate main conclusion and provide next steps:

"To sum up, Horror movie viewership has **increased by 17%** over the past month due our **sponsorship program**. The data suggests that **expanding the sponsorship program** to other genres could lead to similar increases across the board."

Answering “So what?”

- Insights are only as valuable as actions they inspire

- **Example:** Horror viewership is up

✓ **Original version:** *"Horror movie viewership has increased by 17% over the past month, the largest month over month increase ever. This viewership increased due to sponsorship deals with top horror game streamers. By expanding the sponsorship program, we can see similar increases across other genres."*

✗ **Just the first part:** *"Horror movie viewership has increased by 17% over the past month. We analyzed the behavior of over 30,000 users."*

- Connects data to impact and action
- Explains how numbers came to be and what company can do to keep expanding this trend.



Data Storytelling Fundamentals

Selecting insights

Communicating insights

- After wrapping up a data analysis project, you'll be tempted to show everything.
- You have:
 - ✓ Done all the work
 - 🔍 Uncovered all the interesting patterns
 - 📊 Built beautiful charts
- If you include **everything** in your final report, your audience is going to feel overwhelmed.

Anchor the problem

- **Challenge:** Decide what insights to include
- **To choose insights:**

 Anchor to the business problem

 Every insight you include should:

- Answer a key question
- Support a decision

 Go over insights and ask:

- Does this insight help solve the problem we're investigating?
- If I removed this information, would it impact the conclusions?

 Consider level of detail:

- Executive and decision-makers:
 - Appreciate more high-level approach that connects with the bottom line
 - Example: Sales or subscriptions
- Technical teams
 - Appreciate more specific insights
 - Can make sense of additional detail

Avoiding information overload

- **Reminder:** Less is more
- **Ask yourself:**
 - What is the most impactful takeaway?
 - What will drive the biggest decisions?
 - If I had to present just three key points, what would they be?

Example:

- Optimize marketing spending for next quarter
- **Insights:**
 1. Customers spend more in December.
 2. Loyalty program members spend 15% more on average
 3. Customers aged 25–34 tend to buy more luxury items.
 4. Shoppers who visit the website more than three times are more likely to make a purchase.
 5. There's a small but noticeable increase in sales on Thursdays.



Data Storytelling Fundamentals

Identifying your
main conclusion

Report

- One of the most common data analytics artifacts
- Combines text and visualizations to tell a data story
- Looks similar to research report or an essay with charts

 In this lesson, you'll go through steps to create a data analytics report:

- ☐ Identifying main conclusion
- ☐ Determining supporting evidence
- ☐ Ordering evidence
- ☐ Designing report
- ☐ Getting feedback

 In the following lesson:

- ☐ Considerations for other artifacts like slide decks and notebooks

Starting from the insights

- Gathered a range of insights that provide different perspectives
- You'll need to:
 - Craft a narrative
 - Brings insights together in meaningful way for stakeholders
- Main conclusion:
 - ❌ Shouldn't just be a single number
 - ✅ Should explicitly connect key insights to definitions of success

Example: Horror movie viewership

- **Task:** Identify whether sponsorship program was successful
- You should be able to:
 - Give a direct answer
→ **Yes**
 - To support your answer:
 - ☐ Define success
 - ☐ Explain why sponsorship program met that definition

Conclusions can have multiple components

- Okay to have more than one component
- **Example:** Define success for sponsorship program in terms of **user viewership** and **sponsor retention**
- This statement of main conclusion:
 - ✓ Directly answers the question without leaving any ambiguity
 - ✓ Prioritizes factors by ordering them based on impact
 - ✓ Provides a foundation for next steps without dictating solution

*"The sponsorship program was a **success** primarily because **viewership in the target genre increased by 17%** compared with a baseline of 2% in all other genres, and secondarily because **partner satisfaction was over 95%**. By expanding the sponsorship program, we can see similar increases across other genres and grow our partner network."*

Checking in with stakeholders

- Use check-ins with stakeholders to make sure you're still working to answer the right question
- Frame around **what you need to finalize your conclusion**
- **Example:** *"I'm working on answering these questions about viewership. Right now, I've found increased watch time for horror movies, and I suspect it's higher than average. To answer your questions about how unique this trend is, I need additional data on customers watching other genres."*

- ✓ Keeps stakeholders informed
- ✓ Ensures you have all info needed before drawing final conclusion



Data Storytelling Fundamentals

Choosing
supporting evidence

Supporting evidence

- **Main conclusion** → answer to core question

💬 “Was the sponsorship program a success?”

- **Supporting evidence** → answers to the subquestions implied by main question

➡ “How many new and existing users clicked the sponsored links?”

💬 “How much watch time was spent on the target genre, for both new and existing users?”

💬 “How much watch time was spent on average across all genres?”

- First question: Identifies number of users impacted by the program

👤 24,000 users

- 10,000 new
- 14,000 returning

- You can:

✅ Compare with results from other initiatives to understand program’s comparative success.

Exploring vs communicating insights

- You'll be tempted to show everything



Done all the work



Uncovered interesting patterns



Built beautiful visualizations

- Mark of a great data analyst is **restraint!**



Anchor to domain problem and audience

- Think of yourself as a curator:



If you include every fact, figure, and insight, audience will feel overwhelmed



Carefully select most important insights and guide your audience in best possible order

Anchor the problem

- **Challenge:** Decide what insights to include
- **To choose insights:**

⚓ Anchor to the business problem

💡 Every insight you include should:

- Answer a key question
- Support a decision

🔍 Go over insights and ask:

- Does this insight help solve the problem we're investigating?
- If I removed this information, would it impact the conclusions?

Example: Investigate increase in horror movie viewership



Average viewership across all genres increased by 2% in the previous month, while horror movie viewership increased by 17%.



12% of users cancelled their subscriptions last month, down from 14% the previous month.

Ask yourself:

- ☐ What is the most impactful takeaway?
- ☐ What will drive the biggest decisions?
- ☐ If I had to present just three key points, what would they be?

Identifying the best evidence

Strong evidence

- Clearly related to problem
- Backed by reliable data
 - Significant p values
 - Large sample sizes
 - Reliable collection
- Unambiguous in its interpretation

Diverse range of evidence

- Likely part of data story
- May include outliers that can't be explained, but suggest interesting cases.
- Communicate findings with nuance
- Help your stakeholders understand uncertainty

Negative evidence

- Things you expected to matter but actually don't.
- **Example:** New users who clicked sponsorship link did not watch horror movies longer
- Remain steward of truth rather than telling audience what they want to hear



Data Storytelling Fundamentals

Ordering evidence

Bottom up approach

- Think of report as a structured argument
 - Leading audience through logical sequence that supports conclusions
- **Approach 1:** Build from the bottom up
 1. Start with smaller insight
 2. Build on smaller insights
 3. End with main conclusion
- Building from a smaller, easy to grasp insight up to the bigger picture
 - ✓ Shows the chain of reasoning
 - ✓ Helps build trust in analysis process

Example: Tell story about horror viewership using two “average” viewers:

- One who discovered through sponsorship
- One who didn't
- Use individual stories to build broader insight:
 - 👤 “This is Kelly. She is a regular user of our platform who watched her first horror movie this month, and spent 117 mins watching it.”
 - 👤 “This is Fajad. He clicked on a sponsorship link to create a new account and watched two horror movies for a combined total of 140 mins.”
 - 📊 “Now, on our platform we have about 14,000 Kellies and 10,000 Fajads.”

Top down approach

- **Approach 2:** Top down approach
 1. Start with overall population
 2. Break population down into subgroups with more details
- Useful when you want to:
 - Show strong overall growth
 - Identify the source of that growth
- Ordering supporting evidence:
 - ☐ Main conclusion always come first
 - ☐ Followed by data context
 - ☐ Wrap up restating main conclusion with recommended next steps

Example: Horror viewership

- “As a whole, users watched a total of 20 million minutes of horror movies last month.”
- “For context, a person who’s 20 million minutes old would have been born in the 1980s. That number is up from 17.1 million minutes the previous month.”
- “Of these minutes, over 8% came from our sponsored links.”



Data Storytelling Fundamentals

Designing your report

Well-structured report

- Includes core components of data story:
 - Main conclusion
 - Supporting evidence
 - Restated conclusion
- In a report, sections often look like:
 - ☐ Title, author, and other metadata
 - ☐ Executive summary
 - ☐ Data sources and methodology
 - ☐ Walk through supporting evidence
 - ☐ Conclusion and recommendations

Goal:

- ☒ Easy to navigate
- ☒ Coherent when read in order

Tips for visual appeal

- ☐ Use a single consistent **font** throughout
- ☐ Organize sections with hierarchical **headings**
 - Make headings stand out with extra spacing or boldface
- ☐ Use a consistent **color scheme**, with only 2-3 colors used throughout
 - Your company may have a set of “brand guidelines” that specify logo, colors, fonts, and other design elements



Data Storytelling Fundamentals

Getting feedback
on your report

Feedback process

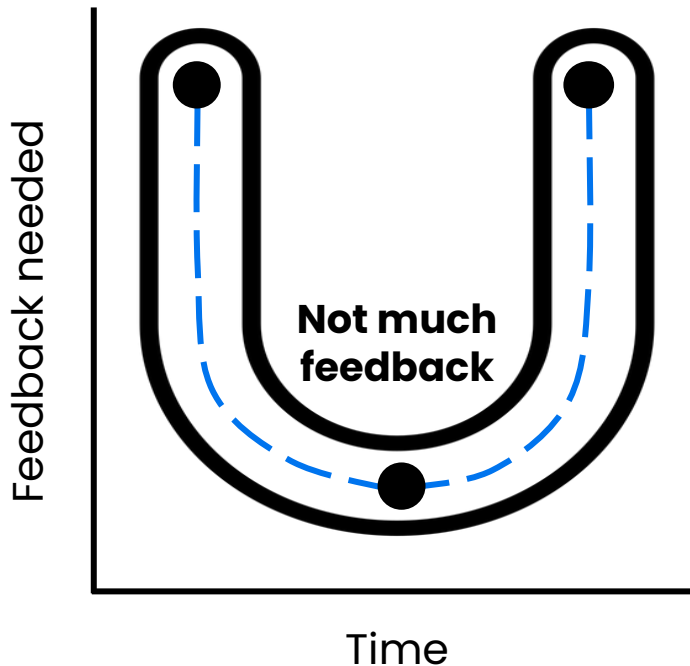
Feedback needed depends on **where you are** in the process.

First starting out:

- Is the structure clear?
- Are the key insights emphasized?
- Does the report follow a logical flow?



Not looking for surface level polish!



Towards the end:

- Clarity of messaging
- Data accuracy
- Presentation polish

How to ask for the right feedback

- Feedback is only as good as the way you ask for it
- **Example:**
 - ❌ Question: "Hey, can you take a look?"
 - ❌ Broad feedback: "Looks great!"
- **Good ways to phrase your request:**
 - ✅ "I'm unsure about the flow between sections two and three. Does the transition work?"
 - ✅ "Does my analysis provide strong enough evidence for my conclusion?"
 - ✅ "Are there areas where my language could be more precise?"

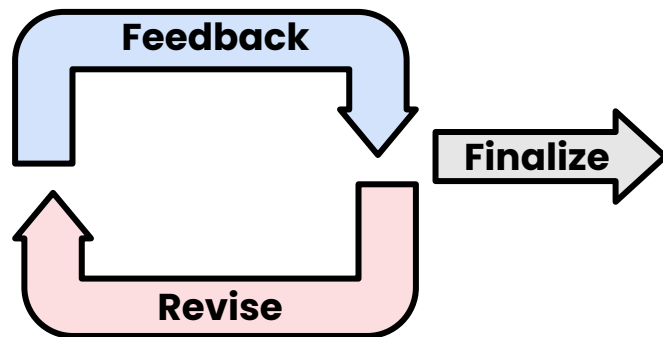
Iterative feedback

- Getting feedback isn't one-and-done
- Back-and-forth process
- Returning to an earlier section to:
 - ☐ Refine evidence
 - ☐ Adjust format based on new insights
- Best reports go through multiple rounds

Round 1: Structure

Round 2: Clarity and messaging

Round 3: Formatting and polish



Using an LLM as a report critic

- If you don't have immediate access to a reviewer, LLM can be first-round critic
 - ❌ **Can't:** replace human feedback
 - ✅ **Can:** provide useful suggestions
- You can ask it to:
 - Evaluate the clarity of your introduction.
 - Check for logical inconsistencies in argument.
 - Suggest improvements in tone and style



Data Storytelling Fundamentals

Creating a memo

Data storytelling formats

- **Toyota:** A3 report
 - Sheet of paper twice as large as typical printer paper
- **Intel:** One-pager
 - Fit all relevant information on a single page
- **Amazon & Netflix:** Popularized the internal memo

Amazon's memo culture

- Jeff Bezos **banned** PowerPoint
- Required every meeting start with reading a six page memo
- Communicate in a data-dense written format while respecting everyone's time
- "Clear writing is clear thinking"

What is a memo?

A memo is:

- A brief, structured document
- Used for:
 - Internal communication
 - Important info
 - Updates
 - Recommendations
- More concise, flexible, and collaborative

Purpose of Memos

- Short and to the point
- Drive decisions
- Shared collaboratively for comments in the document

Structuring your memo

To: (Your primary audience)
From: (Your name or team)
Date: (When the memo is being issued)
And Subject: (A one-line summary of the key insight)

Over the past quarter, our subscriber churn rate increased by 10%, primarily due to low early engagement and price sensitivity.

- **Engagement matters:** Users who watch less than 2 hours per week in their first three months are 3x more likely to unsubscribe.
- **Price is a factor:** 50% of users who left cite pricing as their top reason for canceling.

To address early engagement, we recommend implementing a personalized onboarding experience and A/B testing different engagement strategies. Additionally, revisiting pricing tiers based on competitor benchmarks may help retain price-sensitive users.

Please review these insights and share any questions or feedback by Friday.

1. Header information

2. Main conclusion

3. Key findings

4. Recommendations

5. Call to action

Feedback on memos

- Seek feedback on memos by:
 -  Asking an LLM
 -  Soliciting feedback from colleagues
- Memos are less formal than reports
- May feel more comfortable sending a draft out earlier and with less polish



Data Storytelling Fundamentals

Creating a notebook

Benefits of Jupyter notebook



Transparency

- See exact steps you took
- Run same code
- Verify results



Reproducibility

- Rerun with updated data
- Get new insights without modifying structure



Integrating narrative with process

- Explains why and how key preprocessing and analysis steps were taken

2. Analyzing User Retention Over Time

Below is a line chart that shows the monthly churn rate (percentage of users who canceled) over the 12 months.

Key Steps:

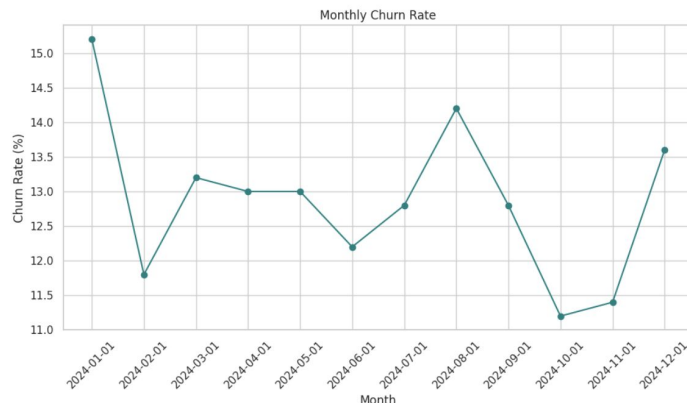
- **Group Data:** We group by **Month** and calculate the churn rate.
- **Visualize Trend:** A line chart makes it easy to observe seasonal or trending patterns.

```
# Convert the 'Churn' column to numbers: 1 for "Yes", 0 for "No"
df['Churn_numeric'] = (df['Churn'] == 'Yes').astype(int)

# Group by 'Month' and calculate the mean churn, then multiply by 100 for the percentage
monthly_data = df.groupby('Month')['Churn_numeric'].mean() * 100

# Reset the index and rename the column
monthly_data = monthly_data.reset_index()
monthly_data.columns = ['Month', 'Churn_Rate']

# Plot the line chart for churn rate over time
plt.plot(monthly_data['Month'], monthly_data['Churn_Rate'], marker='o', color='teal')
plt.title('Monthly Churn Rate')
plt.xlabel('Month')
plt.ylabel('Churn Rate (%)')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```



When to use a notebook

- Choose when communicating with a technical, process-focused audience:
 - ✓ Data team
 - ✓ Engineering team
 - ✗ To non-technical stakeholder, code becomes distracting visual clutter.
- Often a polished version of original notebook with a more defined structure.
- To make a notebook effective:
 - Structure it in a way that makes analysis clear and engaging
 - Add headings, explanations, images, and equations



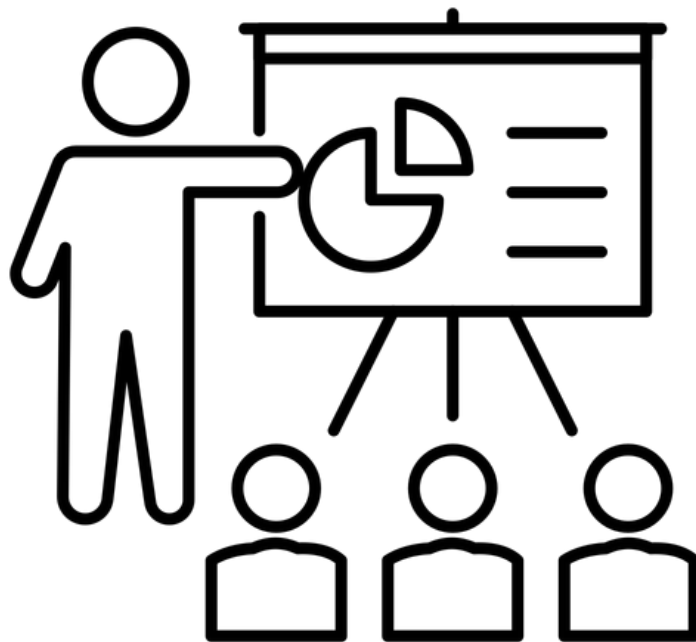
Data Storytelling Fundamentals

Create a slide deck

Why use a slide deck?

When designed and delivered well, slide decks offer several advantages:

- **Controlled delivery**
 - Dictate flow of information so audience stays on track
- **Visual storytelling**
 - Simplify complex data
 - Needs fewer written words
- **Real-time clarifications**
 - Adjust immediately, emphasize key points, or answer questions on spot



Structuring your slide deck

A strong slide deck follows a clear and engaging structure:

1. Title slide
2. Main conclusion slide
3. Context slide
4. Key insights
5. Recommendations

Title

- Clear and concise
- Follows brand guidelines
- Welcome audience and set expectations
- **Example:**
"Thank you for being here! Today, we'll explore the key insights from our latest analysis on subscriber churn, including the concrete steps we can take to improve retention. Let's get started."



Main conclusion

- Answers key business question
- Minimal with just enough information to set stage
- No need for long paragraphs

- Past quarter: **10% increase** in subscriber churn
- Two key factors are driving this trend:
 - Low early engagement
 - Price sensitivity

Context

- Explains more about the data your analysis is grounded in
- This slide explains the:
 - ☐ Population
 - ☐ Sample
 - ☐ Timeline
 - ☐ Key metrics involved

- 1,000 users
- Past 1 year
- Key measures:
 - Watch time
 - Number of sessions
 - Survey results

Key insights

- Highlight most important findings
- Will likely be several of these as you build up narrative that supports main conclusion

Key Insights

- Low engagement in the first three months leads to higher churn.
- 50% of churned users cite pricing concerns.

Recommendations

- Propose data-driven solutions
- Ask for thoughts and feedback to open up a discussion:

"I'd love to hear your thoughts. What questions can I answer about our process, conclusion, and recommendations?"

Recommendations

- Implement a personalized onboarding experience.
- Test new pricing tiers based on competitor benchmarks.

How LLMs can help



Generate speakers notes:

- Generate concise, engaging speakers notes for each slide



Refining slide text

- Make bullet points clearer or more persuasive



Adapting for different audiences

- Tailor slides for executives, technical teams, or marketing professionals



Don't expect LLM to do all the work



Wording might feel a little robotic



Refine into something that sounds natural and engaging



Data Storytelling Fundamentals

Creating appendices

Why appendices matter

- Useful for sharing information that is not critical to the main data story
- Opportunity to provide depth for interested audience members
- Supporting materials that don't need to be in spotlight but are still important
- Strengthens report by building credibility
 - Stakeholders will trust your methods more strongly
- Artifact stays concise

Example: Customer churn

- **Report states:** Low engagement and price sensitivity are biggest drivers of cancellations
- **Appendix:** Data from the survey responses and your code analysis

What belongs in an appendix?

Include:

- ✓ Raw data summaries
- ✓ Methodology and assumptions
- ✓ Detailed calculations or statistical tests
- ✓ Additional charts or tables
- ✓ Glossary

Doesn't belong:

- Anything that's redundant, trivial, or distracts from your core message
- **Example:** Location impacts churn rate
 - Company does not have capability to ship different experiences to different regions
 - May decide not to include finding



Data Storytelling Fundamentals

Documentation best
practices

Documentation

- Supplementary information that accompanies your final report
 - Explains your methodology
-
- ✓ Helps with both transparency and reproducibility
 - ✓ Allows your stakeholders to explore key parameters of your analysis
 - ✓ Allows others in the future to replicate work without additional clarification

Key aspects of documentation

- ☐ Why analysis was conducted and who the audience was
- ☐ Key findings, including how results should and shouldn't be interpreted
- ☐ Exact data sources with instructions for how to access data as relevant
- ☐ Timeframe represented by data, date of access, and the last updated date
- ☐ Author names and contact information
- ☐ Version number of the artifact
- ☐ Version numbers for all Python libraries
- ☐ Well-commented Python and SQL code

Versioning

- Maintaining clear records of authorship, changes, and updates
- Every document should:
 -  Identify author or team responsible
 -  Date it was created
 -  Date of the last update
 -  Assign a version number (e.g. v1, v2)

- **Changelog:**

-  Record of revisions made
-  Quick reference to understand what has been updated and why
-  Keeping a well-maintained history:
 - Enhances credibility
 - Prevents confusion

Consistency

- Applying standardized conventions to data products
- You should standardize:
 - Naming for files
 - Tables
 - Variables
- If working within an organization:
 - Follow established documentation guidelines
- Writing style:
 - Use same terminology across documents to:
 - Prevent misunderstandings
 - Reinforce key concepts
 - **Example:** "Customer churn"
 - ➖ "Customer dropoff"
 - ➖ "User decline"